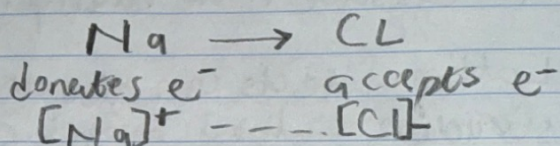
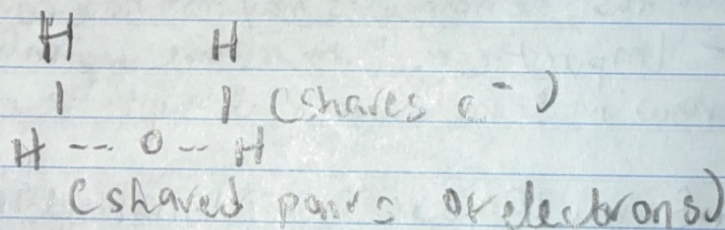


CHE 230 Homework - No Tables

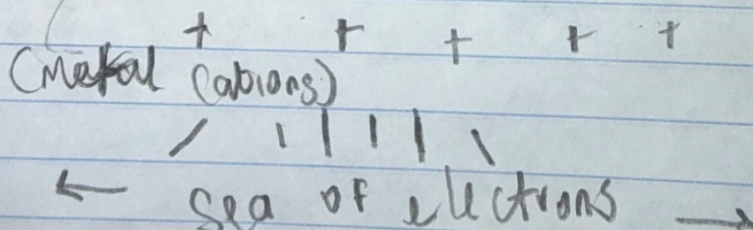
1. Graphite and diamond are both made of carbon atom but they differ in their bonding and structure
2. Ionic bond is formed when one atom gives up one or more electrons to another atom. The atom become oppositely charged ions and the electrostatic attraction between them holds them together



Covalent Bond in this two atoms share electrons to achieve stability. The shared electrons allow both atoms to fill their outer electron shell



Metallic Bond electron are shared freely between atom forming a sea of electron that move freely



3 $APF = \frac{4 \times V \text{ of atoms in unit cell}}{V \text{ of unit cell}}$

a) $V = \frac{4}{2} \pi r^3$
 $V_{\text{atom}} = \frac{4}{3} \pi (0.143 \times 10^{-9})^3$

b) $a = \frac{4r}{\sqrt{2}}$

$\rho = \frac{\text{Mass of unit cell}}{\text{Volume of unit cell}}$

4 a) Solubility: Defect can create disordered regions in the crystal lattice which can cause the material to dissolve more easily.

b) Stability: Lattice defects can also destabilize the crystal structure. They might create sites for the compound to break down or interact with other substances reducing overall stability of the drug.

c - Cooling rate: If the cooling rate during crystallization is too fast the molecules may not have enough time to align.
 - Impurities: If there are impurities in the solvent or raw materials can disrupt the crystallization.

C. 1. Semiconductor Manufacturing
 2. Metal Alloys

5 Corner points

(0, 0, 0)
(1, 0, 0)
(0, 1, 0)
(0, 0, 1)
(1, 1, 0)
(1, 0, 1)
(0, 1, 1)
(1, 1, 1)

✓ 5

Center point

(0.5, 0.5, 0.5)

6

$$n_v = N \cdot e^{-Q_v/KT}$$

$$T(K) = 800 + 273.15 = 1073.15 K$$

$$N = \frac{P \cdot N_A}{A}$$

$$P = 9.5 \text{ g/cm}^3 = 9.5 \times 10^6 \text{ g/m}^3$$

$$N = \frac{(9.5 \times 10^6) \cdot (6.022 \times 10^{23})}{107.9}$$

$$N = 5.31 \times 10^{28} \text{ atoms/m}^3$$

$$\frac{Q_v}{KT} = 1.10$$

$$\frac{Q_v}{KT} = \frac{(8.617 \times 10^{-5}) \cdot 1073.15}{0.0927} = 1.10 = 11.86$$

✓ 6

$$n_v = 8.0 \times 10^{22} \text{ vacancies/m}^3$$

$$\dot{m} = \frac{D \cdot A \cdot (C_1 - C_2)}{L}$$

$$\dot{m} = \frac{(1.0 \times 10^{-8}) \cdot (0.20) \cdot (2.4 - 0.6)}{5 \times 10^{-3}}$$

$$C_1 - C_2 = 2.4 - 0.6 = 1.8 \text{ kg/m}^3$$

$$m = 3.6 \times 10^{-9}$$

$$\frac{3.6 \times 10^{-9}}{5 \times 10^{-3}} = 7.2 \times 10^{-7} \text{ kg/s}$$

$$\dot{M} = 7.2 \times 10^{-7} \times 3600 \text{ s/h}$$

$$\dot{m} = 2.59 \times 10^{-2} \text{ kg/h}$$

$$C(x,t) = C_s - (C_s - C_0) \cdot \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$D = \frac{x^2}{4t \left[\text{erf}^{-1}\left(\frac{C_s - C(x,t)}{C_s - C_0}\right) \right]^2}$$

$$D = D_0 \cdot e^{-Q_d/RT}$$

$$T = \frac{Q_d}{R \ln(D_0/D)}$$

$$4.0 \text{ mm}$$

$$\text{ERF} = 0.99999984583$$

$$C(x,t) = C_s - (C_s - C_0) \cdot \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$D = \frac{x^2}{4t \left[\text{erf}^{-1}\left(\frac{C_s - C(x,t)}{C_s - C_0}\right) \right]^2}$$

$$\frac{1.0 - 0.35}{1.0 - 0.20} = 0.8125$$

$$D = 3.16 \times 10^{-11} \text{ m}^2/\text{s}$$

$$b) T =$$

$$\frac{148,000}{8.314 \cdot \ln\left(\frac{2.3 \times 10^{-5}}{3.16 \times 10^{-11}}\right)} = 13.59$$

$$T = \frac{148,000}{8.314 \times 13.59} = 1319 \text{ K}$$

$$8.314 \times 13.59 = 113.1 \text{ K}$$